

Chapter 3

Operators and Quantum Dynamics

Lecture 7 introduced the Schrödinger equation, energy eigenstates, stationary states, and the free particle. We now make the distinction between time evolution and measurement precise, explain why Schrödinger evolution is unitary, and write the general solution in the energy basis.

Questions.

- What is the difference between applying an operator and measuring an observable?
- What does it mean for time evolution to be unitary?
- Why does a Hermitian Hamiltonian generate unitary evolution?
- How does the continuity equation express local probability conservation?
- How does a general state evolve in an energy basis?

Lecture roadmap. We first review the measurement postulates and separate measurement updates from Schrödinger evolution. We then study unitarity, local probability conservation, and the general energy-basis solution.

Lecture 8: Unitary Evolution

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Review from Lecture 7

The Schrödinger equation

For one particle in one dimension, the time-dependent Schrödinger equation is

$$i\hbar \frac{\partial \psi(x, t)}{\partial t} = \hat{H} \psi(x, t), \quad (3.1)$$

where the Hamiltonian is

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x). \quad (3.2)$$

The Hamiltonian is the total-energy operator: the first term is kinetic energy and $V(x)$ is the potential-energy operator in the position representation.

Energy eigenstates and stationary states

For a time-independent Hamiltonian, an energy eigenfunction satisfies

$$\hat{H}\phi_E(x) = E\phi_E(x). \quad (3.3)$$

If the initial state is $\phi_E(x)$, the Schrödinger equation gives

$$\boxed{\psi_E(x, t) = \phi_E(x)e^{-iEt/\hbar}}. \quad (3.4)$$

The time-dependent factor is a phase, so

$$|\psi_E(x, t)|^2 = |\phi_E(x)|^2. \quad (3.5)$$

Thus an energy eigenstate is called a *stationary state*. The wave function itself changes by a phase, but its position probability density does not change with time.

The free particle

For $V(x) = 0$,

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}. \quad (3.6)$$

The fixed-energy equation has the general solution

$$\phi_E(x) = Ae^{ikx} + Be^{-ikx}, \quad p = \pm\hbar k, \quad E = \frac{\hbar^2 k^2}{2m}. \quad (3.7)$$

The two terms have opposite momenta but the same kinetic energy. A single momentum eigenfunction is often written as

$$\phi_k(x) = \frac{1}{\sqrt{2\pi}} e^{ikx}, \quad \hat{p}\phi_k = \hbar k\phi_k. \quad (3.8)$$

The factor $1/\sqrt{2\pi}$ is chosen for delta normalization; a plane wave is not an ordinary square-normalizable state.

Applying an operator is not measuring an observable

An operator is a mathematical rule that maps one state or function to another. For example,

$$\hat{H}\psi(x) \quad (3.9)$$

is another function, not a measurement result. No apparatus has interacted with the system, and no eigenvalue has been recorded.

Even if

$$\hat{H}\phi_E = E\phi_E, \quad (3.10)$$

no measurement has occurred. This equation says that ϕ_E is a state with definite energy E . If the energy is then measured, the result E occurs with probability one.

For an observable A , the eigenvalue equation is

$$\hat{A}\phi_n = a_n\phi_n. \quad (3.11)$$

The eigenvalues a_n are the possible ideal-measurement outcomes. If the state is expanded as

$$\psi = \sum_n c_n \phi_n, \quad (3.12)$$

then, for a nondegenerate eigenvalue, an ideal measurement gives

$$P(a_n) = |c_n|^2, \quad \psi \longrightarrow \phi_n \quad \text{when } a_n \text{ is recorded.} \quad (3.13)$$

The measurement process includes three distinct ingredients:

1. a physical interaction between the system and a measuring apparatus;
2. a recorded eigenvalue a_n ;
3. a state update to the corresponding eigenstate or eigenspace.

Therefore,

$$\boxed{\text{eigenvalue equation} \neq \text{measurement.}} \quad (3.14)$$

3.10 Two state-update rules

The first-pass formalism has two different rules for changing a quantum state.

3.10.1 Between measurements: Schrödinger evolution

Between measurements, the state evolves according to

$$\psi(x, t_2) = U(t_2, t_1) \psi(x, t_1). \quad (3.15)$$

For a time-independent Hamiltonian,

$$\boxed{U(t_2, t_1) = e^{-i\hat{H}(t_2-t_1)/\hbar}.} \quad (3.16)$$

This evolution is linear, deterministic, continuous, and unitary. In particular, once $\hat{H}$ and $\psi(x, t_1)$ are specified, the later state is determined.

3.10.2 During an ideal measurement

If a measurement records a_n , the ideal state update is

$$\psi \longrightarrow \phi_n. \quad (3.17)$$

The outcome is probabilistic, and the update is generally not reversible from the final state alone. The post-measurement state supplies a new initial condition for the next interval of Schrödinger evolution.

3.10.3 A measurement followed by evolution

Suppose an energy measurement at $t = t_*$ gives E_n . Immediately after the measurement,

$$\psi(x, t_*) = \phi_n(x). \quad (3.18)$$

For $t > t_*$, the new initial condition gives

$$\boxed{\psi(x, t) = \phi_n(x) e^{-iE_n(t-t_*)/\hbar}.} \quad (3.19)$$

The measurement selects a new state; Schrödinger evolution then propagates that state. One should not continue the pre-measurement evolution through the measurement without changing the initial condition.

3.11 Unitarity and probability conservation

3.11.1 Definition of a unitary operator

An operator U is unitary if it preserves inner products:

$$\int_{-\infty}^{\infty} [Uf(x)]^* [Ug(x)] \, dx = \int_{-\infty}^{\infty} f^*(x)g(x) \, dx. \quad (3.20)$$

Equivalently,

$$\boxed{U^\dagger U = UU^\dagger = \hat{1}.} \quad (3.21)$$

Thus

$$U^{-1} = U^\dagger, \quad (3.22)$$

so unitary evolution is reversible.

3.11.2 Norm preservation

Setting $f = g = \psi$ in the inner-product relation gives

$$\int_{-\infty}^{\infty} |U\psi(x)|^2 \, dx = \int_{-\infty}^{\infty} |\psi(x)|^2 \, dx. \quad (3.23)$$

Therefore,

$$\boxed{\int_{-\infty}^{\infty} |\psi(x, t_2)|^2 \, dx = \int_{-\infty}^{\infty} |\psi(x, t_1)|^2 \, dx.} \quad (3.24)$$

The norm of the wave function is preserved, so a normalized state remains normalized and total probability remains one.

3.11.3 Why a Hermitian Hamiltonian gives unitary evolution

For a time-independent Hamiltonian, write

$$U = e^{-i\hat{H}\Delta t/\hbar}. \quad (3.25)$$

Because the Hamiltonian represents the observable energy, it is Hermitian:

$$\hat{H}^\dagger = \hat{H}. \quad (3.26)$$

Consequently,

$$U^\dagger = e^{+i\hat{H}\Delta t/\hbar}, \quad (3.27)$$

and

$$U^\dagger U = e^{+i\hat{H}\Delta t/\hbar} e^{-i\hat{H}\Delta t/\hbar} = \hat{1}. \quad (3.28)$$

The logical chain is

$$\boxed{\text{Hermitian Hamiltonian} \Rightarrow \text{unitary evolution} \Rightarrow \text{probability conservation.}} \quad (3.29)$$

3.12 Local probability conservation

Unitarity gives global probability conservation. The Schrödinger equation also gives a local conservation law describing how probability flows through space. Define the probability density

$$\rho(x, t) = |\psi(x, t)|^2. \quad (3.30)$$

The Schrödinger equation and its complex conjugate are

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi, \quad (3.31)$$

$$-i\hbar \frac{\partial \psi^*}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi^*}{\partial x^2} + V\psi^*. \quad (3.32)$$

Using

$$\frac{\partial \rho}{\partial t} = \psi^* \frac{\partial \psi}{\partial t} + \psi \frac{\partial \psi^*}{\partial t}, \quad (3.33)$$

one obtains the continuity equation

$$\boxed{\frac{\partial \rho}{\partial t} + \frac{\partial j}{\partial x} = 0,} \quad (3.34)$$

where the probability current is

$$\boxed{j(x, t) = \frac{\hbar}{2mi} \left(\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x} \right).} \quad (3.35)$$

For an interval $a < x < b$, integration of the continuity equation gives

$$\frac{d}{dt} \int_a^b \rho(x, t) dx = j(a, t) - j(b, t). \quad (3.36)$$

The probability inside the interval changes only through its boundaries: the change is inflow minus outflow. For the whole real line,

$$\frac{d}{dt} \int_{-\infty}^{\infty} |\psi|^2 dx = -[j(+\infty) - j(-\infty)]. \quad (3.37)$$

For a normalizable state with vanishing boundary current,

$$\boxed{\frac{d}{dt} \int_{-\infty}^{\infty} |\psi|^2 dx = 0.} \quad (3.38)$$

3.13 General solution in the energy basis

3.13.1 Expansion in energy eigenstates

Suppose the energy eigenfunctions satisfy

$$\hat{H}\phi_n(x) = E_n\phi_n(x). \quad (3.39)$$

Each eigenfunction evolves as

$$\phi_n(x) \longrightarrow \phi_n(x)e^{-iE_nt/\hbar}. \quad (3.40)$$

An arbitrary initial state can be expanded as

$$\psi(x, 0) = \sum_n c_n \phi_n(x). \quad (3.41)$$

By linearity, each component evolves independently, giving

$$\psi(x, t) = \sum_n c_n \phi_n(x) e^{-iE_nt/\hbar}. \quad (3.42)$$

For an orthonormal discrete energy basis,

$$c_n = \int_{-\infty}^{\infty} \phi_n^*(x) \psi(x, 0) \, dx, \quad P(E_n) = |c_n|^2. \quad (3.43)$$

The coefficients are determined by the initial state. Their squared moduli are the probabilities of the possible energy-measurement outcomes.

Lecture 8 summary

- Applying an operator is a mathematical operation. A measurement produces a recorded eigenvalue and updates the state.
- Between measurements, Schrödinger evolution is linear, deterministic, and unitary. A measurement supplies a new initial condition.
- Unitarity preserves inner products, norms, and total probability:

$$U^\dagger U = \hat{1}.$$

- For a Hermitian Hamiltonian,

$$\hat{H}^\dagger = \hat{H} \implies U = e^{-i\hat{H}t/\hbar} \text{ is unitary.}$$

- The continuity equation expresses local probability conservation:

$$\frac{\partial \rho}{\partial t} + \frac{\partial j}{\partial x} = 0.$$

- A general state evolves in the energy basis as

$$\psi(x, t) = \sum_n c_n \phi_n(x) e^{-iE_nt/\hbar}.$$

Suggested reading. Griffiths and Schroeter, *Introduction to Quantum Mechanics*, 3rd ed., sections on time evolution, stationary states, and expansion in energy eigenstates. Review Lecture 7 on the Schrödinger equation and energy eigenstates.